

Stat 337 - Homework 2: TWO-Way ANOVA

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Due: Wednesday, March 4, 2026 by 11:59pm in Gradescope

Critiquing Published Research

A learning outcome of this course is to:

Demonstrate an understanding of statistical results as used in journal articles, and discuss scope of inference in the context of authentic research questions and datasets.

To answer the following questions, you will need to read:

The Surprising Power of a Click Requirement: How Click Requirements and Warnings Affect Users' Willingness to Disclose Personal Information Epstein R, Zankich VR (2022). *The surprising power of a click requirement: How click requirements and warnings affect users' willingness to disclose personal information*. **PLOS ONE**, 17(2): e0263097. <https://doi.org/10.1371/journal.pone.0263097> (Available on the course webpage as *The surprising power of a click requirement.pdf*)

1. (4 pts) Based on the **3.1 Participants** section, can we **generalize** the results from the study **to all people worldwide**? Be specific in your response to this question and include context provided in the paper.
 - **The results cannot be generalized to all people worldwide. This is because the participants were all selected from the Amazon Mechanical Turk website.**
2. (2 pts) In section **3.1 Participants**, the authors report group sizes of 304, 304, 304, 305, and 333, then use SPSS's "Random sample of cases" to select **300 per group**. Given that discarding data reduces the information available, and that randomly trimming induces another layer of sampling variability (a different draw could yield slightly different results), **what potential benefit do you think motivated the authors to sub-sample from their observations and induce equal sample sizes across groups ($n_{jk} = n$)?** Be specific.
 - **A potential benefit of sub-sampling from their observations to create equal sample sizes across groups is to create a balanced design, which will simplify anova. It will also reduce the tests sensitivity to violations of homogeneity and variance**
3. (4 pts) In **4.1 Analysis of variance**, the authors report a summary of statistical findings of **$F(1,1196) = 8.6794$, $p < 0.01$** for a test of the **interaction** between **warning strength** and **click requirement** on the true mean number of "YES" selections per. (Note: they do not include the control treatment in this two-way

ANOVA analysis). Using the parameter definitions below, **write the full theoretical ANOVA model that allowed them to test for the interaction.**

$$Y_{ijk} = \alpha + \tau_j + \gamma_k + \omega_{jk} + \epsilon_{ijk} \text{ where } \epsilon_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2)$$

Parameter definitions

- i : observation index (participant in treatment j, k) $i = 1, 2, \dots, 300$
 - j : level of warning strength $j = 1, 2$ (weak, strong)
 - k : level of click requirement $k = 1, 2$ (weak, strong)
 - Y_{ijk} : response for observation i (number of “YES” admissions)
 - α : the true mean number of “YES” for the baseline
 - τ_j : the true deviation from the true mean for weak warning, weak click to the j^{th} level of warning strength
 - γ_k : the true deviation from the true mean for weak warning, weak click k^{th} level of click requirement
 - ω_{jk} : the true interaction adjustment for the j^{th} level of warning strength and the k^{th} level of click requirement
 - ϵ_{ijk} : the true difference in the true mean response for group j, k and the i th observation in that group ($\epsilon_{ijk} = y_{ijk} - \mu_{jk}$).
4. (4 pts) In plain English, and in proper context, explain what an interaction between **warning strength** and **click requirement** means for the outcome (number of “YES” selections).
 - **The effect of click requirement is dependent on whether the warning strength is strong or weak.**
 5. (4 pts) Can they claim that **warning strength** *causes* the mean number of “YES” responses to change, and that the amount it changes is dependent on the level of **click requirement**? **Yes, since participants were randomly assigned to levels of warning strength and click requirement. Therefore the effect of warning strength on yes responses does depend on the level of click requirement**
 6. (4 pts) Given that the authors find evidence that there is an interaction between **warning strength** and **click requirement** and (presumably) to include the control group in the analysis, the authors create a single variable they call “Group” with levels (G1, G2, G3, G4, G5). What treatment (i.e., combination of levels of the two factors of interest) is associated with each of the levels of “Group”? Define them below.
 - **G1 - Control(no warning strength, no click requirement)**
 - **G2 - weak warning strength, no click requirement**
 - **G3 - weak warning strength, click requirement**
 - **G4 - strong warning strength, no click requirement**

- **G5 - strong warning strength, click warning strength**
7. (2 pts) The paper runs a one-way ANOVA using “Group” as the single factor of interest and then reports multiple **pairwise comparisons** but does **not** state whether any **multiple-comparisons adjustment** was used. **Why does this matter?**
 - **This matters because when there are multiple pairwise comparisons each test has a chance to produce a Type 1 error. Multiple comparison adjustments correct for inflated risk making results more accurate.**
 8. (4 pts) Based the following summary from **4.2 Control group versus experimental groups**: “G1 = G2 < G3 = G4 = G5”, describe what the cld output would provide (assuming the multiple comparisons were done appropriately). That is, make up letters to go with each of the treatment groups.
 - **G1 = a**
 - **G2 = a**
 - **G3 = b**
 - **G4 = b**
 - **G5 = b**
 9. (4 pts) The paper uses the data to address quite a few more research questions than they elude to in the introduction which states their main objective as, “*What kind of privacy-related warnings might cause internet users to be more cautious about what they reveal online?*” Revisit Section 2.8 of your textbook on the “reproducibility crisis.” Do the secondary questions and analyses given you any cause for concern?
 - **Yes they are a cause for concern, this is because findings from unplanned exploratory analyses are less likely to be accurate because they were not a part of pre-established hypothesis or objectives**

Exploratory data analysis (EDA)

While the authors released a subset of their data, we do not have access to the full dataset, likely because they respected participants’ privacy preferences. As a result, we cannot fully reproduce their reported results. Instead, we will use the available data to explore a potential **interaction between Education and Condition** on the **mean number of “YES” selections**.

We’ll be using a subset of the original data file which includes the following variables:

YES: The number of “YES” admissions a participant gave on a 25-item sensitive-behaviors checklist (0–25). Higher values indicate more disclosures (*treating as a numeric count to follow what the authors did*). [Note: this is actually a binomial response (bounded count), so we are assuming a continuous response here when the response is actually categorical. This is okay under certain conditions, and you can learn more about this by taking STAT 411 and STAT 412!]

condname: The experimental **Condition** shown before the checklist, combining warning type and click requirement.

- TOSA warning - click required (strong warning + must click)
- TOSA warning - NO click required (strong warning, no click)
- Beginning warning - NO click required (weaker brief warning, no click)
- No TOSA or warning (control)

education: Participant's highest education level (collapsed for analysis):

- HS_or_less (no HS / HS diploma)
- Associate (2-year degree)
- Bachelors (4-year college)
- Graduate (Master's/Doctorate)

id: A simple row identifier for the observation. (*index; not used in analysis*)

10. (4 pts) The following code loads the dataset, generates summary statistics, and creates a set of plots. Use these plots to assess whether there is visual evidence of an interaction between condname and education. In your response, **reference specific features of the interaction plot**, in context.

- **The plot shows evidence of a potential interaction between condname and education because the four condition lines are non-parralel accross education levels.**

```
## 'data.frame': 998 obs. of 4 variables:
## $ id : int 1 2 3 4 5 6 7 8 9 10 ...
## $ YES : int 1 15 1 14 14 8 10 15 11 12 ...
## $ condname : chr "No TOSA or warning" "No TOSA or warning" "No TOSA or w
arning" "No TOSA or warning" ...
## $ education: chr "HS_or_less" "Graduate" "HS_or_less" "Graduate" ...

## Rows: 998
## Columns: 4
## $ id <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1
7, 1...
## $ YES <int> 1, 15, 1, 14, 14, 8, 10, 15, 11, 12, 19, 15, 6, 15, 6, 1
5, 1...
## $ condname <chr> "No TOSA or warning", "No TOSA or warning", "No TOSA or
warn...
## $ education <chr> "HS_or_less", "Graduate", "HS_or_less", "Graduate", "HS_
or_1..."

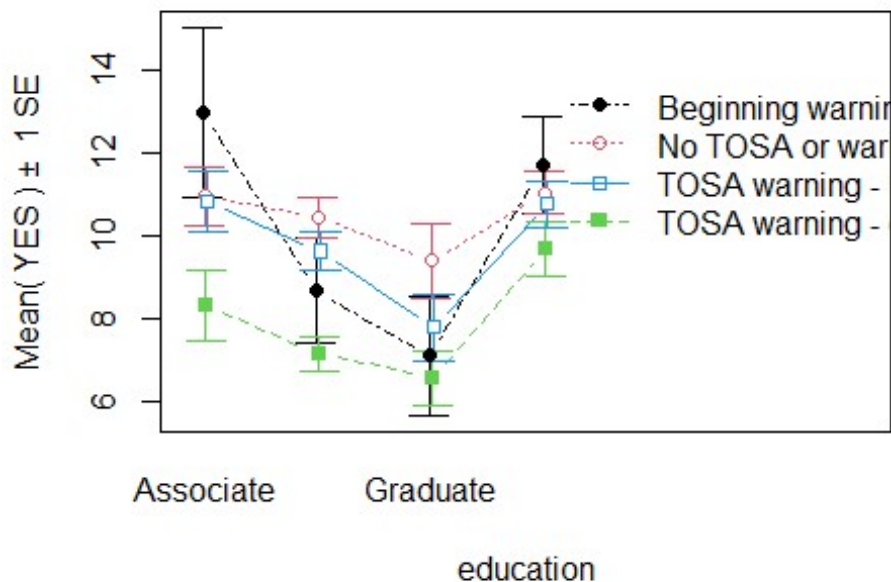
## condname.education min Q1 median Q3
max
## 1 Beginning warning - NO click required.Associate 4 8.25 13.0 17.75
24
## 2 No TOSA or warning.Associate 0 7.00 12.0 15.00
21
## 3 TOSA warning - click required.Associate 0 4.00 9.0 12.00
19
```

```

## 4      TOSA warning - NO click required.Associate  0 8.00  11.0 14.00
21
## 5  Beginning warning - NO click required.Bachelors  0 1.50  11.0 12.50
18
## 6      No TOSA or warning.Bachelors  0 7.00  11.0 15.00
21
## 7      TOSA warning - click required.Bachelors  0 2.00   7.0 11.25
20
## 8      TOSA warning - NO click required.Bachelors  0 6.00  10.0 14.00
20
## 9  Beginning warning - NO click required.Graduate  0 4.00   8.0 10.00
13
## 10     No TOSA or warning.Graduate  2 5.50   9.0 13.00
18
## 11     TOSA warning - click required.Graduate  0 2.00   5.5 11.00
18
## 12     TOSA warning - NO click required.Graduate  0 3.75   8.0 11.00
18
## 13  Beginning warning - NO click required.HS_or_less  2 8.00  12.0 16.00
19
## 14     No TOSA or warning.HS_or_less  0 8.00  11.0 15.00
23
## 15     TOSA warning - click required.HS_or_less  0 6.00  10.0 14.00
21
## 16     TOSA warning - NO click required.HS_or_less  0 7.00  10.5 14.00
23
##      mean      sd    n missing
## 1  13.000000  6.497863  10      0
## 2  10.983051  5.393136  59      0
## 3   8.325000  5.455590  40      0
## 4  10.844444  5.058546  45      0
## 5   8.695652  6.063563  23      0
## 6  10.457944  4.966454 107      0
## 7   7.160256  5.324922 156      0
## 8   9.652778  5.416008 144      0
## 9   7.111111  4.314060   9      0
## 10  9.407407  4.601499  27      0
## 11  6.575758  5.285861  66      0
## 12  7.812500  4.568034  32      0
## 13 11.705882  4.831727  17      0
## 14 11.053097  5.507853 113      0
## 15  9.703125  5.359103  64      0
## 16 10.790698  5.222387  86      0

```

interaction plot of YES based on education and condi



11. (4 pts) Write out the hypotheses for the test for an “interaction,” both using parameters (i.e., in notation) and in words, in the context of the problem. Be sure to define all indices that you use (i.e., i, j, k). (*Hints for math mode: $\alpha, \tau_j, \gamma_k, \omega_{jk}, \neq, =$*)
- $H_0: \omega_{jk} = 0$ for all j, k
 - $H_a: \omega_{jk} \neq 0$ for at least on j, k
 - H_0 : After accounting for the main effect of condition and education level, there is no interaction between warning condition and education level on the mean number of yes responses.
 - H_a : After accounting for the main effect of condition and education level, there is an interaction between warning condition and education level on the mean number of yes responses.

Analysis and diagnostics

Write code to fit the 2-way ANOVA “interaction model” and provide an ANOVA table.

```
# fit the interaction model
fit_int <- lm(YES ~ condname * education, data = dat)
# run the ANOVA
anova(fit_int)

## Analysis of Variance Table
##
```

```
## Response: YES
##               Df Sum Sq Mean Sq F value    Pr(>F)
## condname       3  1583.7   527.90 18.9003 6.371e-12
## education      3   759.8   253.26  9.0674 6.342e-06
## condname:education  9   220.8    24.54  0.8785  0.5439
## Residuals     982 27427.9    27.93
```

12. (4 pts) Write an evidence statement and conclusion *in the context of the problem* for the hypotheses associated with the interaction term (those outlined in Q11). Be sure to include the value of the test statistic, its assumed distribution under the null hypothesis, and a p-value in your evidence statement. Your conclusion should state the interaction in the context of the problem (not as a vague “interaction”!)

- **Under the null, the test statistic follows a $f(9, 982)$ distribution. The test statistic was $F = 0.8785$ and the p-value of 0.5439 provides little to no evidence against the null. We fail to reject the null. There is little to no evidence that the effect of condname on the mean number of “YES” responses is the same across all levels of education**

13. (1 pt) We now decide to fit the additive model that includes condition (condname) and education (education) but not their interaction. Choose the appropriate ANOVA function to test the hypotheses for each factor’s impact on the mean response, after accounting for the other factor in the model in the ANOVA table. (Delete the code you do not need!)

Run the ANOVA on this additive model to test the null hypothesis:

H_0 : The true mean number of Yes selections by MTurk workers is the same across conditions, after accounting for education level.

```
# fit the additive model, notice the '+' between Factor A and Factor B
fit_yes_add <- lm(YES ~ condname + education, data = dat)
```

```
# ANOVA tables
```

```
Anova(fit_yes_add, type = "II")
```

```
## Anova Table (Type II tests)
```

```
##
```

```
## Response: YES
```

```
##               Sum Sq Df F value    Pr(>F)
```

```
## condname    1116.9    3 13.3447 1.524e-08
```

```
## education    759.8    3  9.0774 6.244e-06
```

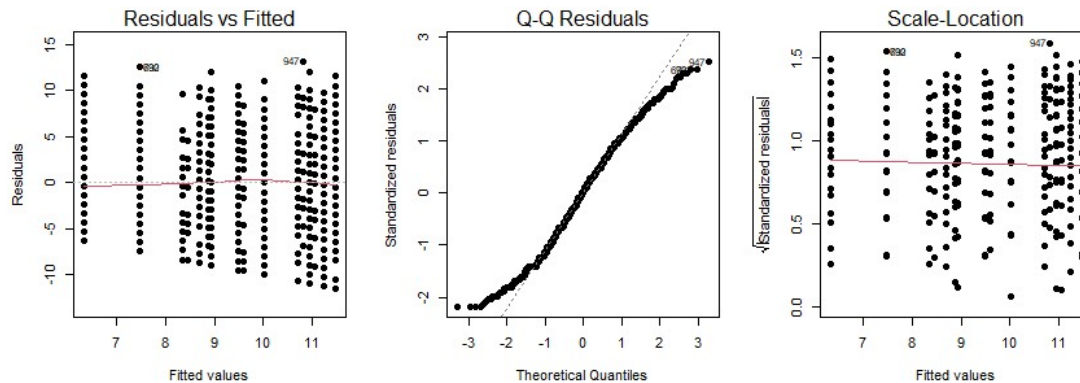
```
## Residuals 27648.8 991
```

Residual Diagnostics

Diagnostic plots are provided below for assessing the equal variance and normality assumptions from the additive two-way anova model.

```
# split plotting window into 1 row, 3 columns
par(mfrow = c(1,3))

# make diagnostic plots
plot(fit_yes_add, which = c(1,2,3), pch = 19)
```



14. (2 pt) Based on the residual diagnostic plots above, select the most appropriate statement:

- A. The pattern of the residuals in the Normal Q-Q plot reveals a heavy-tailed distribution, giving us strong evidence against the normality assumption.
- B. The pattern of the residuals in the Normal Q-Q plot reveals a light-tailed distribution, giving us weak evidence against the normality assumption.
- C. The pattern of the residuals in the Normal Q-Q plot reveals a right-skewed distribution, giving us moderate evidence against the normality assumption.
- D. The pattern of the residuals in the Normal Q-Q plot reveals a left-skewed distribution, giving us moderate evidence against the normality assumption.

• **B**

15. (4 pts) Examine the residual diagnostic plots and **assess the strength of the evidence against the equal variance assumption**. In your response, make sure to:

- State whether the evidence against equal variance is weak, moderate, or strong.
- Reference which plot(s) you used.
- Discuss the specific features of the plot(s) that support your assessment.
- **There is moderate evidence of a violation in equal variance since there is a slight fanning in the residuals vs. fitted plot**

16. (4 pts) Regardless of your answer above, use the summary output generated in the next code chunk to find the **estimated mean number of disclosures (“YES”)** for

participants in the “**Strong warning with click requirement**” condition who have an **Associate** education level. Use proper notation to denote the estimated mean for this group and show all of your work. *You may use latex in the .Rmd file itself, or insert a picture of a neatly handwritten model into your final document. Do this as the final step (add it to the Word file after knitting), because each re-knit will regenerate the document.*

```
# obtain estimated parameter values for additive model
```

```
summary(fit_yes_add)$coef
```

```
##              Estimate Std. Error    t value
## (Intercept)    10.8070718   0.7885217  13.7054850
## condnameNo TOSA or warning    0.4162728   0.7522312   0.5533841
## condnameTOSA warning - click required  -2.1201688   0.7488841  -2.8311042
## condnameTOSA warning - NO click required -0.1120534   0.7515035  -0.1491056
## educationBachelors    -1.2062792   0.4989420  -2.4176743
## educationGraduate    -2.3423145   0.6305843  -3.7145146
## educationHS_or_less    0.2475302   0.5300624   0.4669832
##              Pr(>|t|)
## (Intercept)    2.811264e-39
## condnameNo TOSA or warning    5.801252e-01
## condnameTOSA warning - click required  4.732601e-03
## condnameTOSA warning - NO click required 8.815007e-01
## educationBachelors    1.579961e-02
## educationGraduate    2.149637e-04
## educationHS_or_less    6.406146e-01
```

- $\hat{\mu}_{Click, Associate} = \hat{\alpha} + \hat{\tau}_{click} + \hat{\gamma}_{Associate}$
- $\mathbf{\hat{\mu}}_{Click, Associate} = 10.807 + (-2.120) + 0 = 8.687$

17. (2 pts) Please list anyone whom you worked on any part of this homework with if their name is not already part of this group’s submission. For tutors, give their name and affiliation (MSC, SmartyCats, etc.). For fellow students, give their name and section number. If you did not utilize any outside resources, simply enter “NA.”

- **NA**